Project Planning & Management

PortfolioGenie — AI-Powered Developer Portfolio Builder

1. Project Overview

PortfolioGenie is a web-based platform designed to help junior developers create and customize a professional online portfolio. The platform guides users through a structured, multi-step form to enter their personal information, skills, and projects. Users can preview their portfolio while editing it, and an AI-assisted feature can help improve their biography and project descriptions.

2. Problem Statement

Junior developers often need a clear and professional way to present their skills and projects to recruiters, but building a portfolio website from scratch can require time, design experience, and front-end development knowledge. PortfolioGenie aims to simplify this process through guided data entry, reusable layouts, live preview, and optional AI writing assistance.

3. Project Aim and Objectives

- Provide a simple workflow for creating a developer portfolio.
- Allow users to enter and update profile details, technical skills, and project information.
- Provide a multi-step form that makes the creation process easier to follow.
- Enable live preview so users can see how their portfolio changes as they edit it.
- Use prompt engineering with an AI service to suggest clearer biographies and project descriptions.
- Follow responsive design and accessibility principles.
- Maintain clean, modular, and easy-to-maintain code.

4. Project Scope

In Scope

- A landing page explaining the platform.
- A guided portfolio creation form.
- Profile, skills, and project sections.
- Portfolio editor and live preview navigation.
- AI-assisted text improvement for the biography and project descriptions.


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- Responsive user interface and basic accessibility support.

Out of Scope for the Initial Version

- A full job application or recruitment system.
- A social network or public discussion platform.
- Complex payment and subscription features.
- Guaranteed automatic publishing to every hosting provider.

5. Target Users and Stakeholders

- Primary users: junior developers and students who need a professional portfolio.
- Secondary stakeholders: recruiters or hiring managers who view published portfolios.
- Project stakeholders: the project team, mentor, and DEPI evaluators.

6. Proposed Technology Stack

The following technologies are based on the PortfolioGenie project description supplied for this assignment. Any additional backend or database technology should be confirmed against the complete official project brief.

Area	Technology / Requirement	Purpose
Frontend	React	Build reusable UI components and application pages.
Language	TypeScript	Add static typing and improve maintainability.
React fundamentals	useState, useEffect	Manage component state and side effects.
Reusable logic	Custom Hooks	Encapsulate logic for the multi-step form.
Navigation	React Router	Navigate between editor and live preview.
UI/UX	Design Thinking, User Personas, Wireframes	Plan a user-centred and accessible experience.
AI	Prompt Engineering + AI service	Improve biographies and project descriptions; support SEO-oriented wording.
Code quality	Clean code and maintainable structure	Make the application easier to test and extend.

7. Work Breakdown Structure

Phase	Main Activities	Deliverable
1. Planning	Confirm goals, scope, roles, and timeline.	Planning document
2. Research	Review comparable portfolio-building solutions.	Literature review
3. Requirements	Document user needs and functional/non-functional requirements.	Requirements specification


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4. UI/UX Design	Create user personas, wireframes, and page flow.	Design artifacts
5. Frontend Development	Build form steps, editor, preview, and responsive layouts.	Working frontend
6. AI Integration	Design prompts and connect the AI feature securely, if supported by the approved brief.	AI writing assistance
7. Testing	Test form validation, navigation, responsiveness, and AI error states.	Test report
8. Final Delivery	Prepare demo, source code, and final documentation.	Final project package

8. Proposed Timeline

The timeline below is a planning proposal and should be adjusted to the team's actual schedule and mentor instructions.

Week	Planned Work
Week 1	Planning, research, and requirements gathering
Week 2	User personas, wireframes, and interface design
Week 3	Multi-step form and portfolio editor
Week 4	Live preview and responsive styling
Week 5	AI prompt design and integration
Week 6	Testing, bug fixing, documentation, and presentation

9. Risks and Mitigation

Risk	Potential Impact	Mitigation
Unclear requirements	Rework and missed features	Confirm scope with the project brief and mentor before implementation.
AI API limits or downtime	AI assistance may be unavailable	Show clear error messages and allow manual editing without AI.
Responsive layout issues	Poor mobile experience	Test common viewport widths throughout development.
Overly large scope	Late delivery	Prioritize the core portfolio workflow before optional enhancements.
Poor input validation	Incorrect or incomplete portfolio data	Validate required fields and provide clear feedback.

10. Success Criteria

- A user can complete the portfolio form and review the entered information.
- The editor and live preview navigation work correctly.
- The interface remains usable on mobile and desktop screens.
- AI suggestions, if enabled, can be accepted, edited, or ignored by the user.

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- The code is organized into reusable components and documented sufficiently for handover.

11. Assumptions

This plan is based on the visible PortfolioGenie brief, which specifies UI/UX practices, React Hooks, Custom Hooks for a multi-step form, React Router, AI prompt engineering, and clean code. The visible brief does not confirm a mandatory backend, database, authentication system, or hosting provider; these should not be treated as official requirements unless stated elsewhere in the complete instructions.

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